

initial battery setup

Issue: didn't account for spark, overpowered for my purposes and much harder to manage with two batteries at once

bus bar



Male EC5 to 8mm ring terminal adapter



EC5 1-female 2-male splitter



Batteries



new battery setup:

- battery -> ec5 female
- ec5 male -> QS8-S female
- QS8-S female-> QS8-S male
- QS8-S female -> ring terminal
- ring terminal -> bus bar

drivetrain escs to bus bar:

- XT90 to ring terminal

flywheel escs to bus bar:

- bare wire to ring terminal
- ring terminal to bus bar

new battery setup

QS8-S connectors.

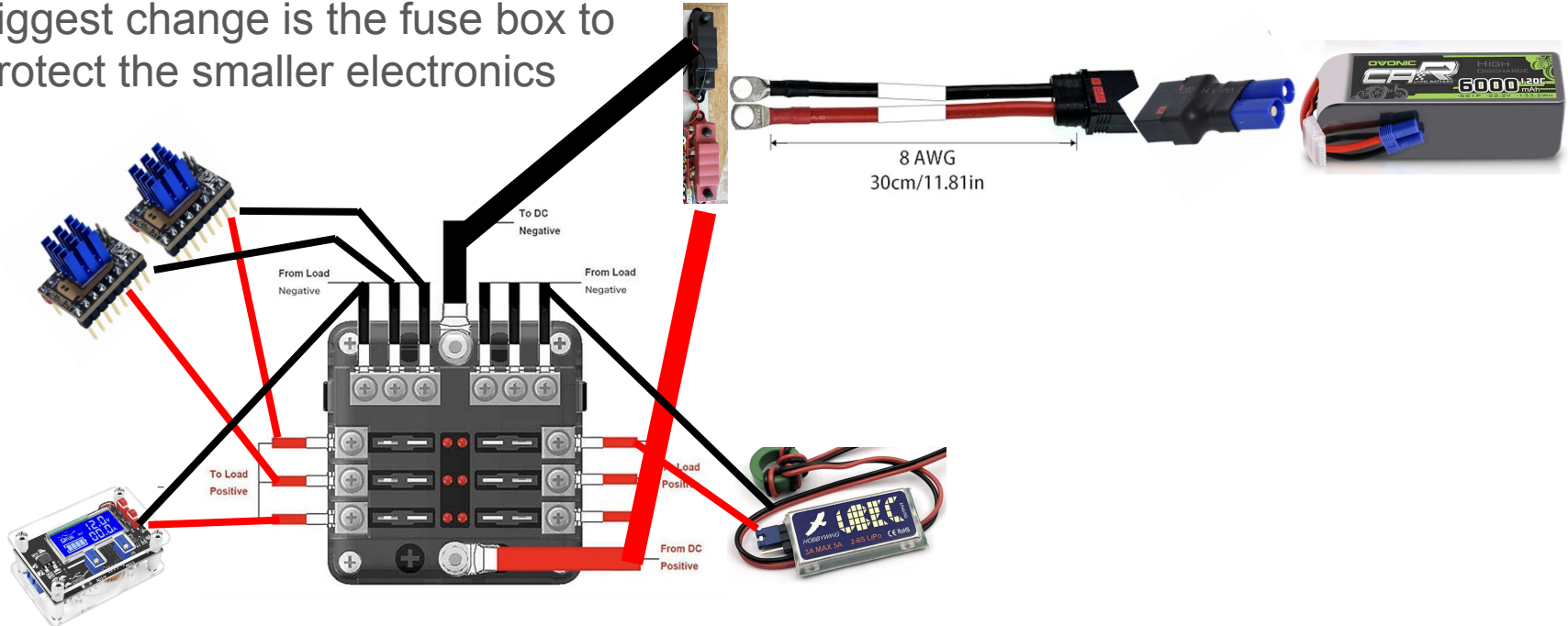
- built-in anti-spark
- rated for up to 300A continuous



only one battery

New power distribution setup

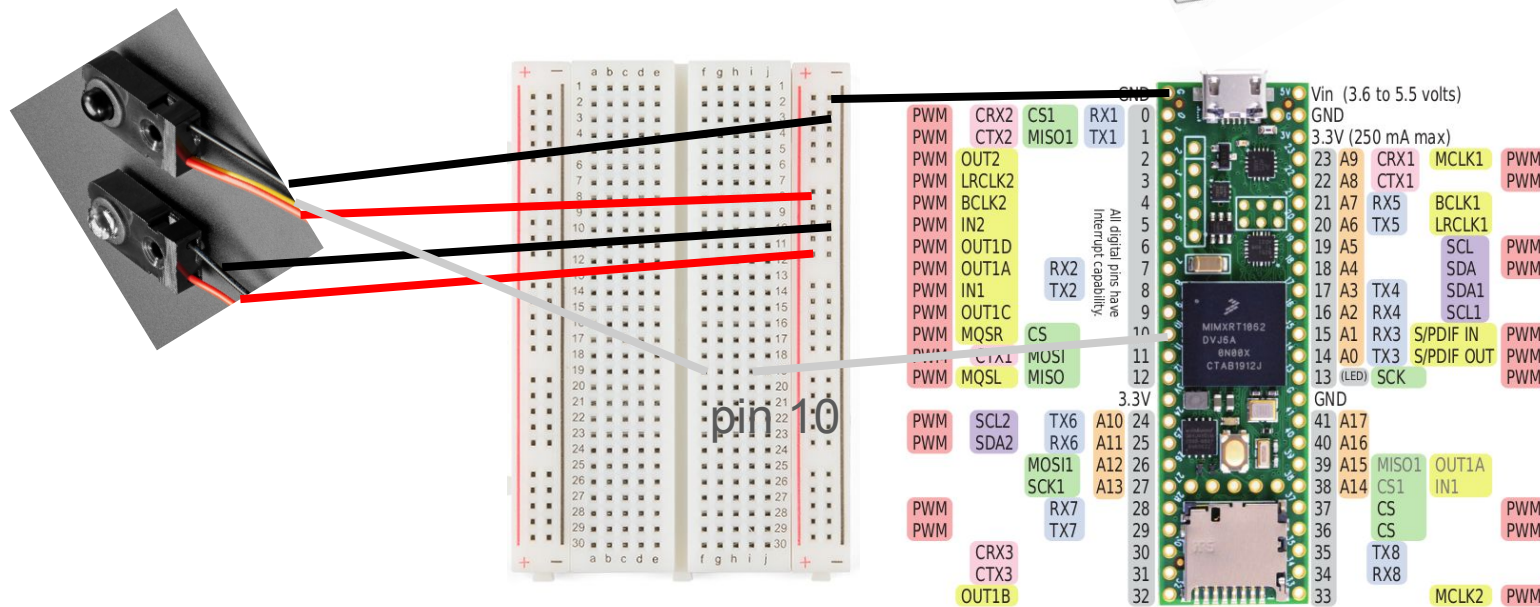
biggest change is the fuse box to protect the smaller electronics



cut the copper trace between V_{in} and V_{usb} on the teensy so that i can have the battery powering the teensy and view the serial output through usb at the same time

note: use protoboard and dupont connectors

eventually use battery power

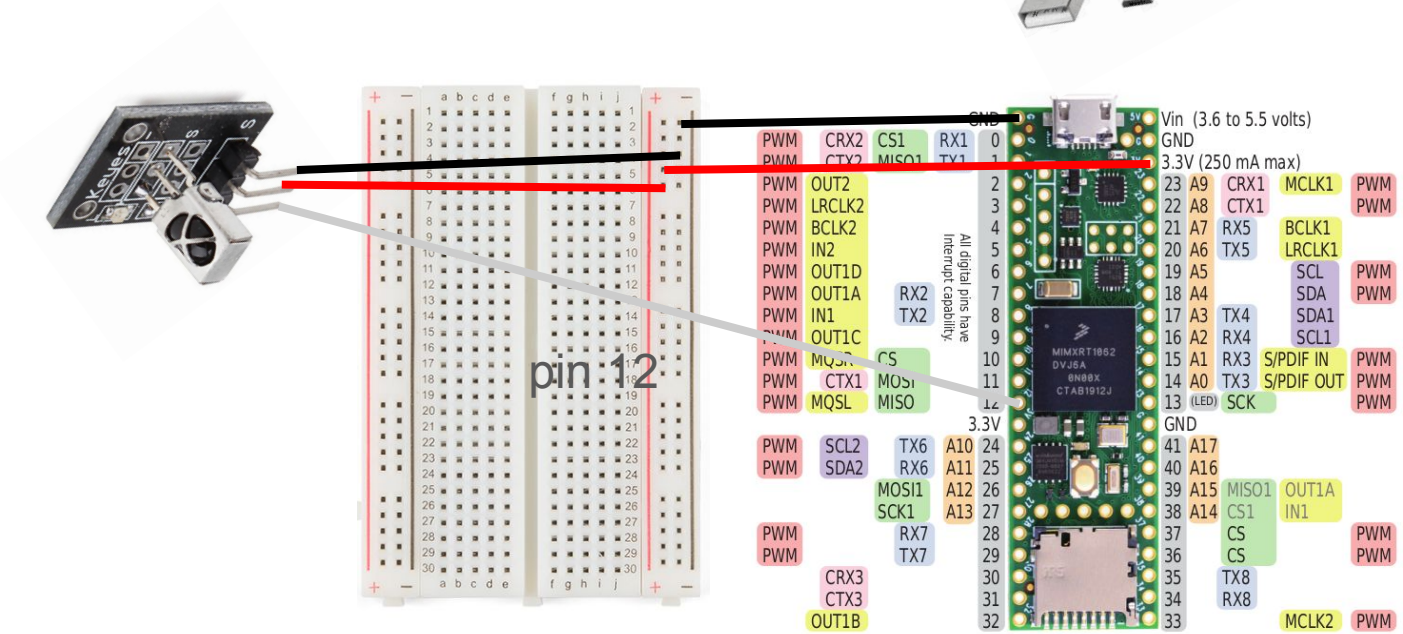


remote control working with the teensy

note: use protoboard and dupont connectors

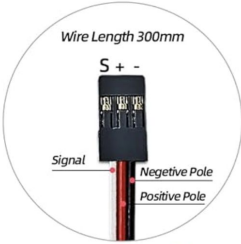
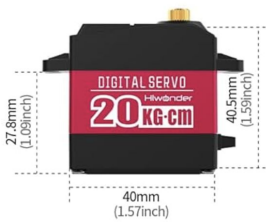
from laptop

eventually use battery power

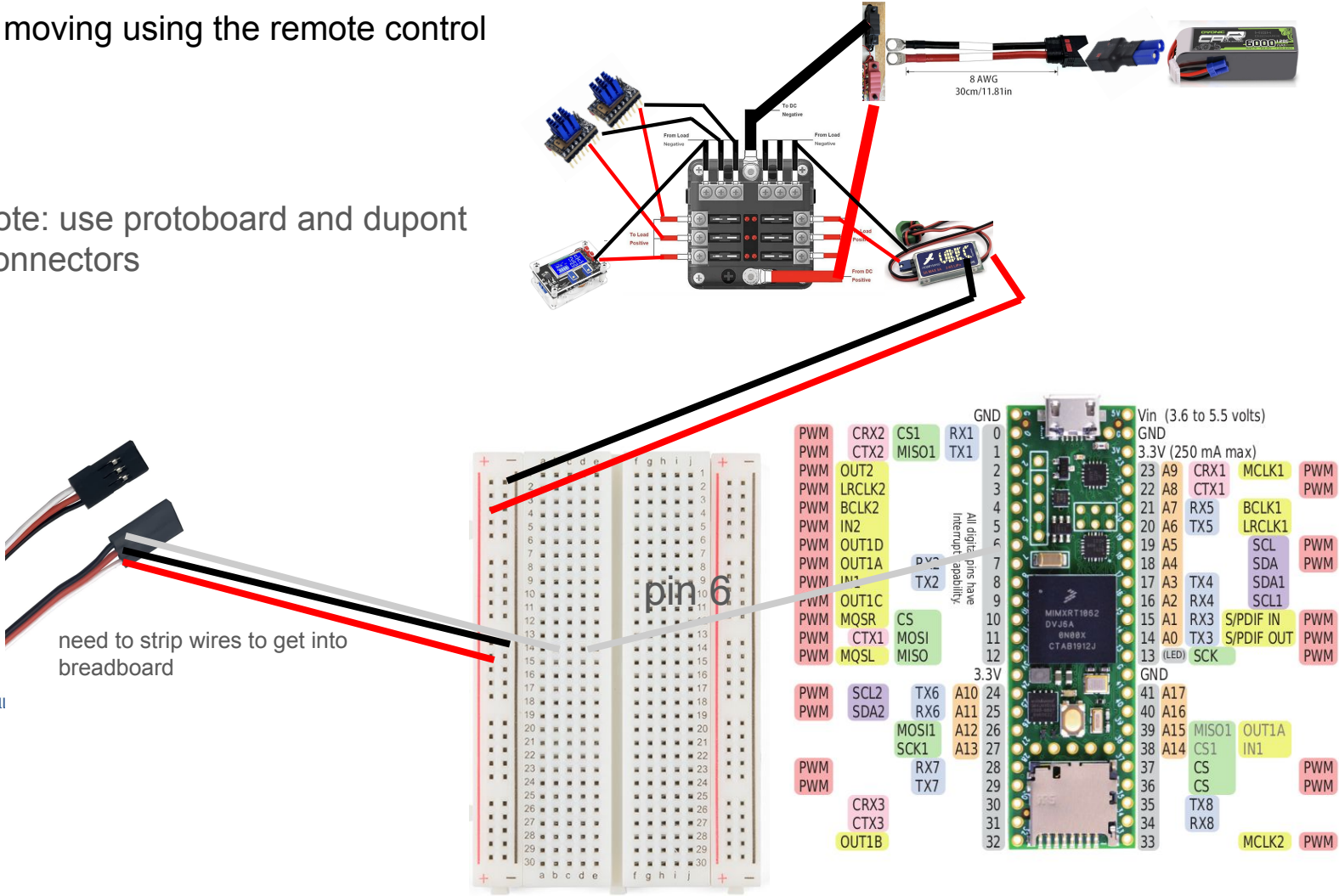


The servo motor moving using the remote control

note: use protoboard and dupont connectors

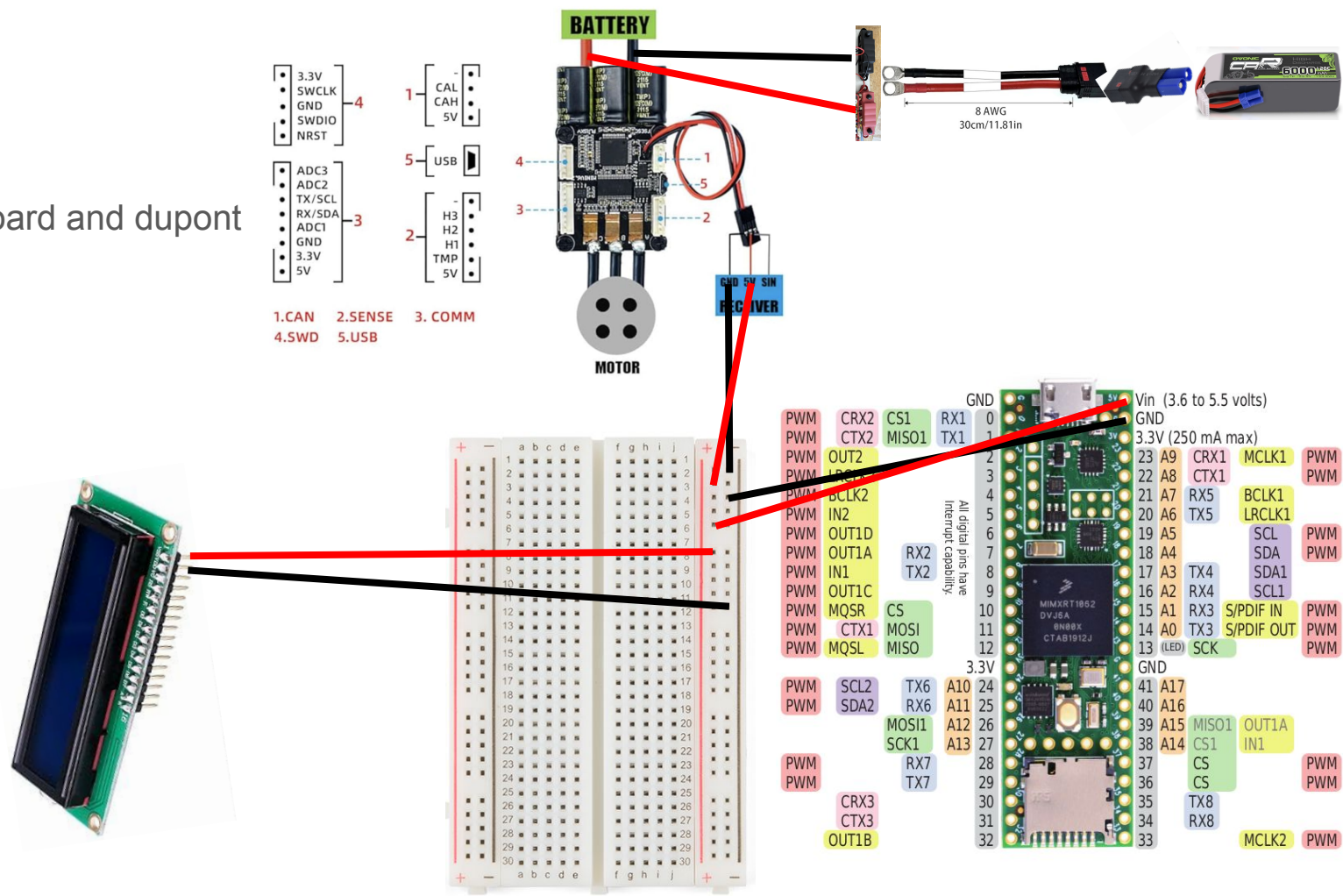


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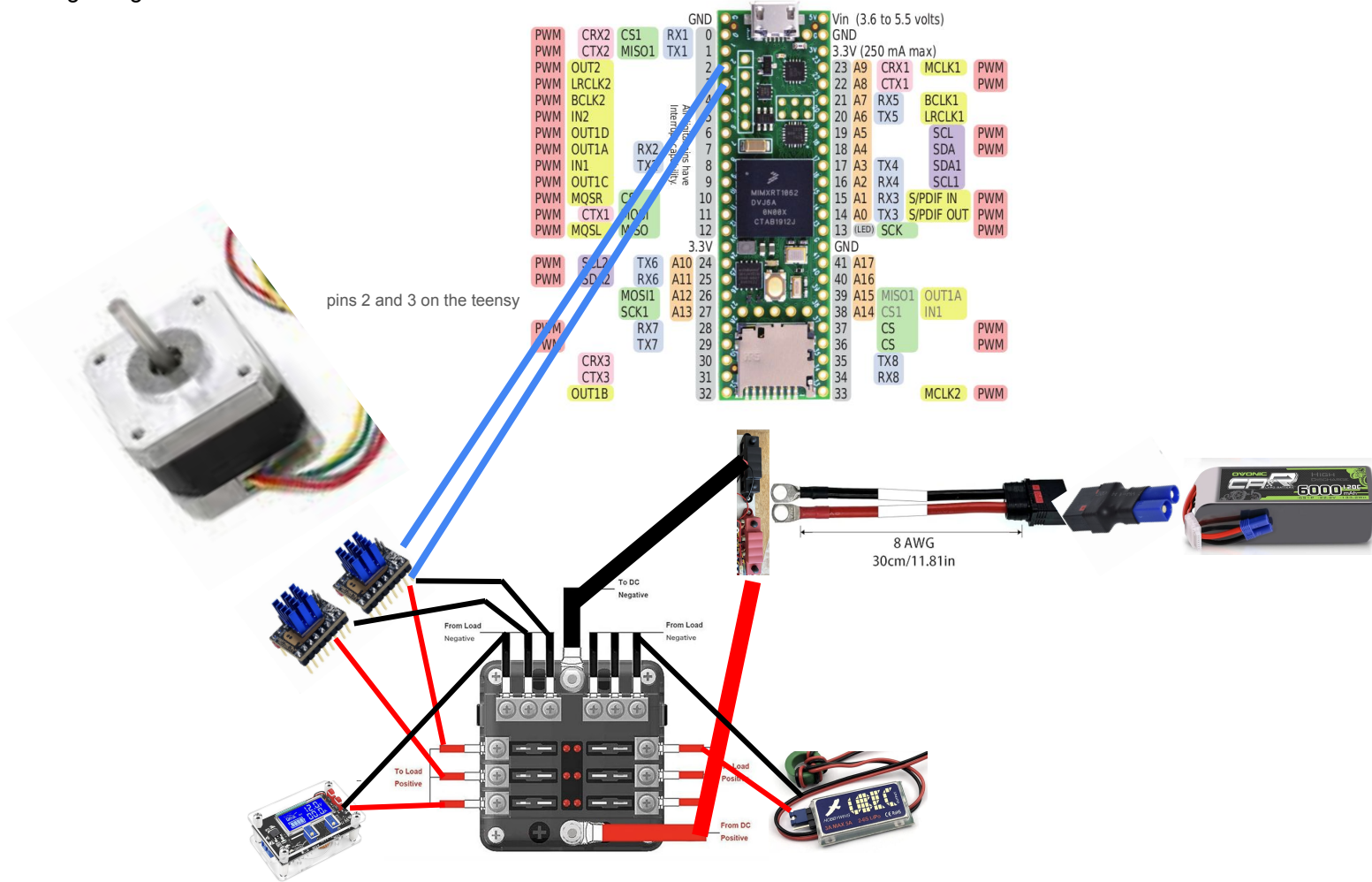


LCD screen display working

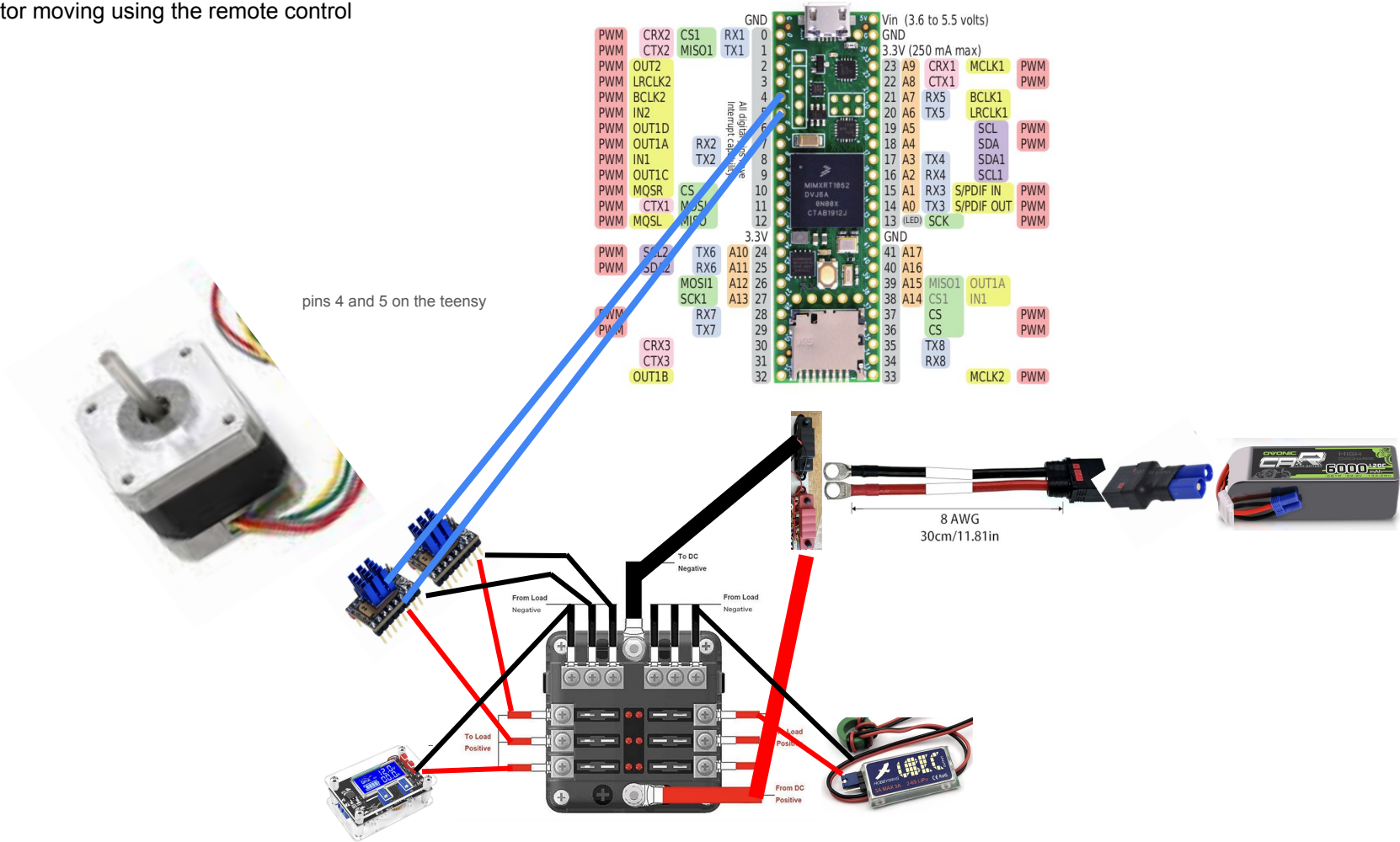
note: use protoboard and dupont connectors



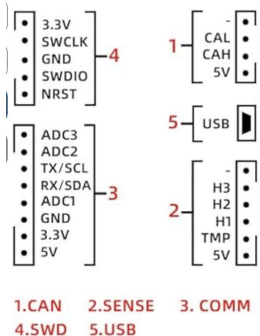
The camera pitch motor moving using the remote control



The staging motor moving using the remote control



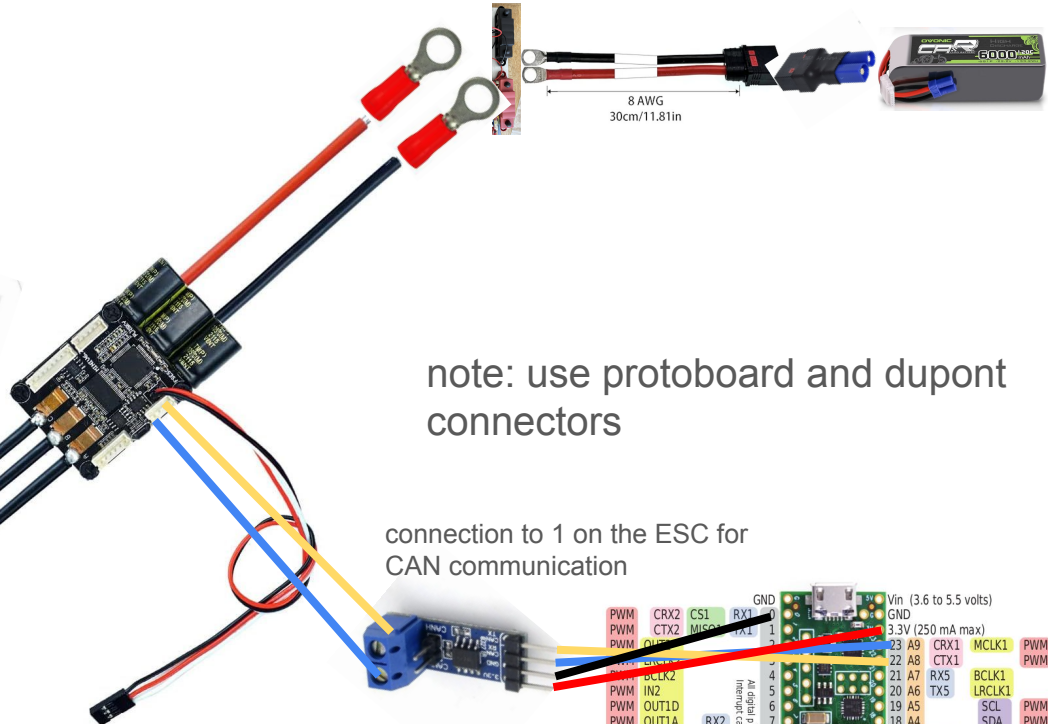
The flywheel motors moving using the remote control



motor jst connectors plug into 2 on the ESC (SENSE for hall effect sensors)

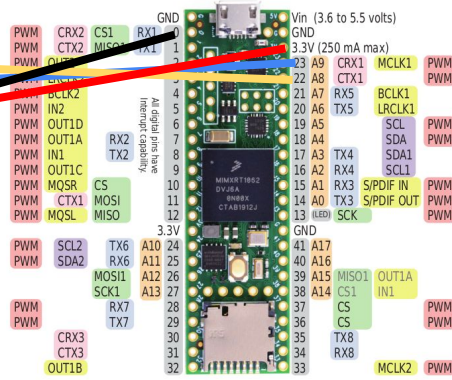


3x 4.0mm gold bullet connectors (female soldered onto ESC's outgoing bare wires).

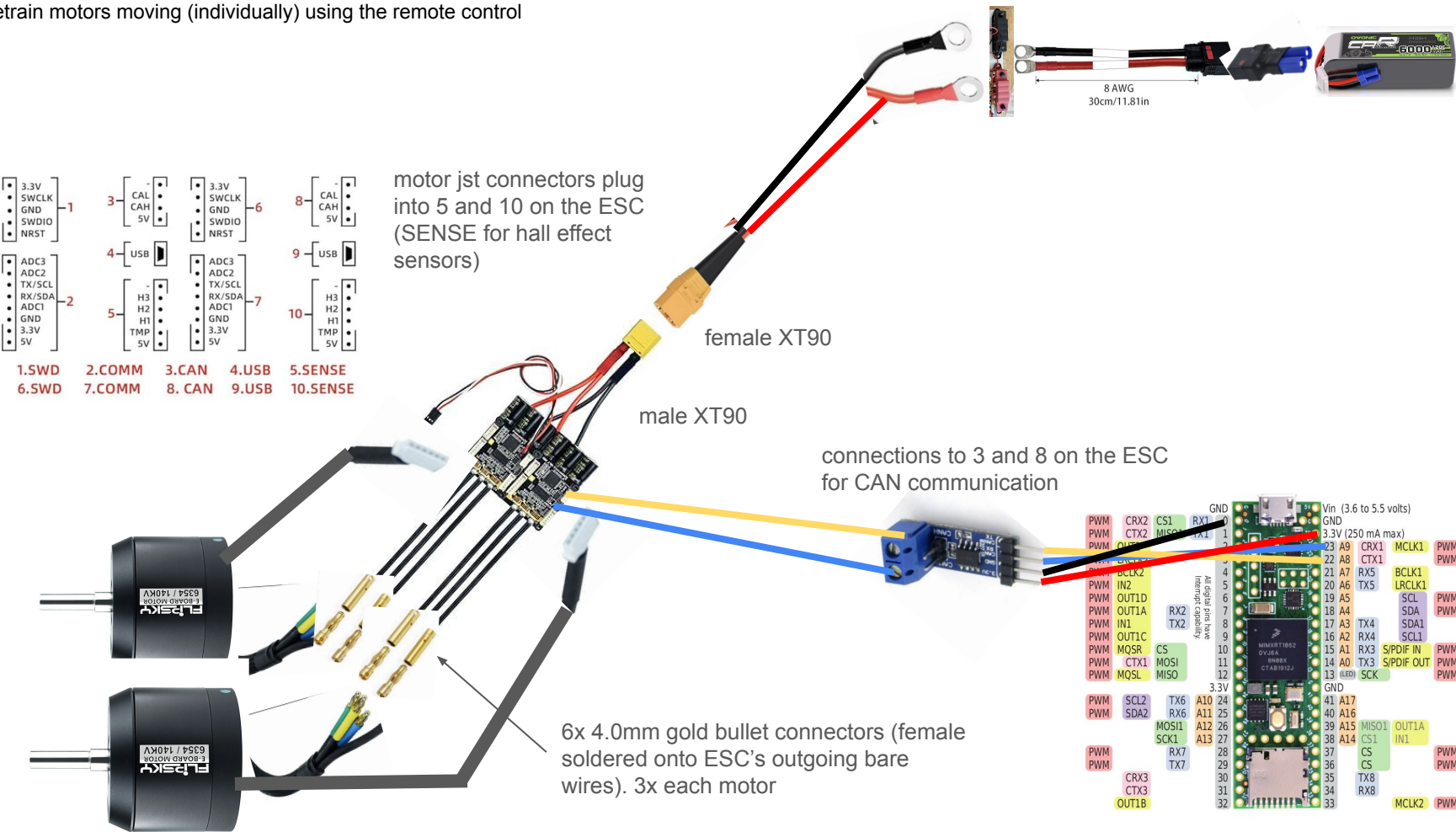


note: use protoboard and dupont connectors

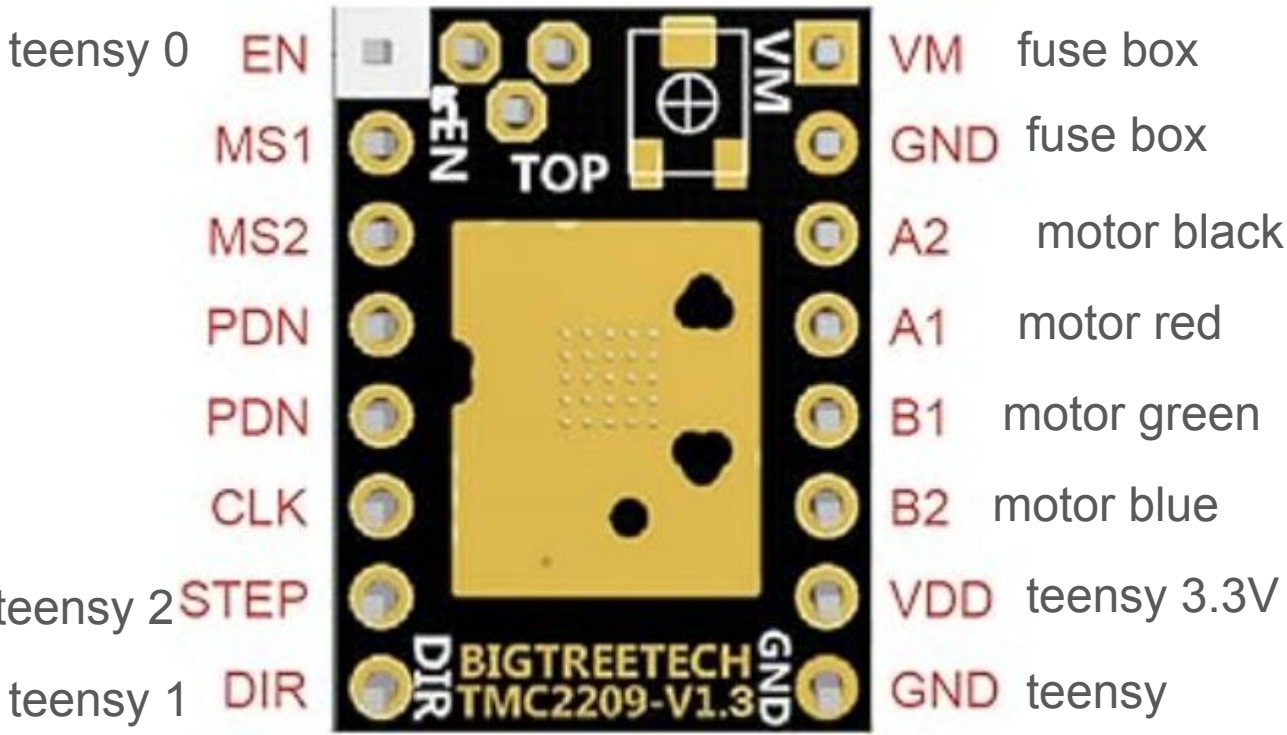
connection to 1 on the ESC for CAN communication



The drivetrain motors moving (individually) using the remote control



stepper motor driver connections (camera pitch controller)



stepper motor driver connections (staging controller)

